Aerospace Information Technology
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Floating Satellite System (FloatSat)

Control System Design

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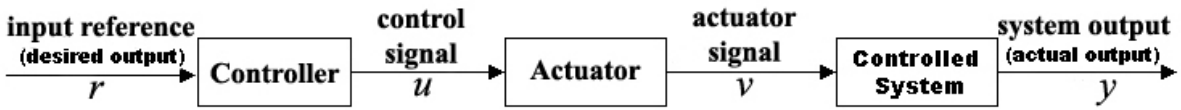
Fundamentals of Control Theory:

• **Control theory** is an interdisciplinary branch of **engineering and mathematics** that deals with the **behavior of dynamical systems**. When a system output needs to track a desired input over time, a controller, that manipulates an actuating signal to the system to obtain the desired effect on the output, is needed.

• **Open Loop Control System (non-feedback control):** The **controller** computes its actuating signal (**control signal**) using only the **desired input** information and no feedback of the system output is needed (**no sensors information**).

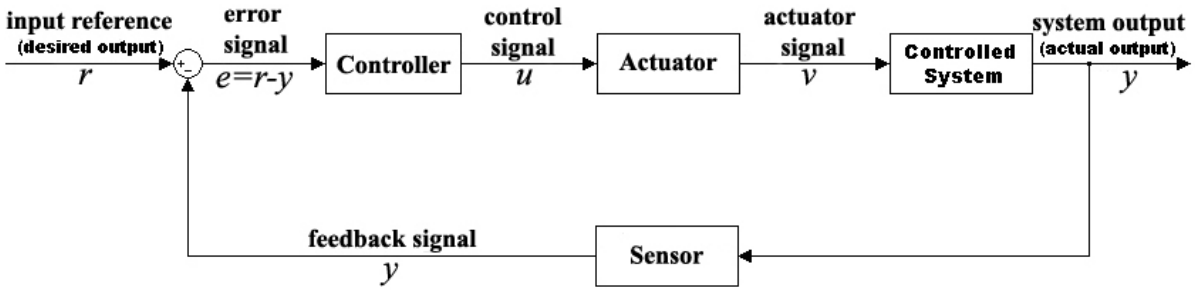


Problems:

• can only counteract against disturbances, for which it has been designed.
• cannot used to stabilize inherently unstable processes.
• sensitive to parameter variations.
• cannot guaranteed performance with model uncertainties.

Fundamentals of Control Theory:

• **Closed Loop Control System (feedback control):** **Sensors** are used to measure actual states of the system (**actual outputs**) and feed these information back to the controller. The **controller** uses these information to **adjust** the **actuating signal** accordingly by **comparing** them to the **desired input** information (**error signal**).



Advantages:

• counteract against unmeasured disturbances.
• can be used to stabilize inherently unstable processes.
• adaptive to parameter variations.
• can guarantee performance with model uncertainties.

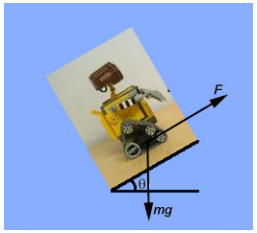
System Modelling and Identification:

In order to achieve a certain desirable performance for a given plant, it is first necessary to **derive a model for a system** that is adequate for controller design and **accurately describe the dynamics of the system**. In general, the goal of **system modelling** is to **determine a mathematical model for a system or a process**.

- **White Box Model:** an analytical model developed by use of "**laws of nature**".

Example: The momentum balance equation of the cruise control system an be written as:

$$m\frac{dv}{dt} + cv = F - mg\sin\theta$$



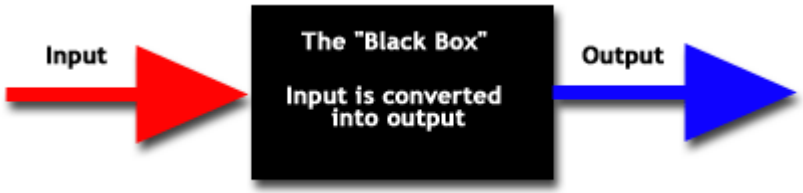
where the term ***cv*** describes the momentum loss due to air resistance and rolling and ***F*** is the force generated by the motor. The retarding force due to the slope of the road (disturbance) should similarly be proportional to the sine of the angle which can approximated to θ.

Problems:

- Analytical models can be overly complex and possibly even impossible to obtain in reasonable time due to the complex nature of many systems and processes.

System Modelling and Identification:

• **Black Box Model:** An experimental model based on an experimentation process known as system Identification. The problem of system identification is determining a system model from the relationship (either in the time or frequency domain) between its input and output using statistical methods from measured data.



– Step response based methods are most commonly used for system identification.
– Simple system identification method involves selecting an appropriate governing differential equation and finding corresponding coefficients.

$$\sum_{i=0}^{n} a_i y^{(i)}(t) = \sum_{j=0}^{m} b_j u^{(j)}(t)$$

where $n \geq m$

System Modelling and Identification:

The systems can be divided based on the analytic equations describing their dynamics into two main mathematical categories: **linear** and **nonlinear**.

- If all the operators in a mathematical model exhibit linearity, the resulting mathematical model is defined as linear. Otherwise, the model is considered to be nonlinear.

– A variety of analysis and design techniques for **linear systems**, include methods like **root locus analysis**, **Nyquist criterion**, **Bode plot**, and **pole placement**, can be used in the design process.

– In **nonlinear systems**, these techniques cannot necessarily be applied directly. Nevertheless, in nonlinear control theory, it is sometimes possible to **linearize** such classes of systems using **linearization techniques**, such as **Jacobian** and **feedback linearization**, in order to be able to apply such linear methods.

Example: The momentum balance equation of the cruisec system

$$m\frac{dv}{dt} + cv = ku - mg\sin\theta$$

-The relationship between the system output (system velocity v) and the system input (motor voltage u) is linear.

-The relationship between the system output (system velocity v) and the disturbance input (the slope of the road θ) is nonlinear and need to be linearized.

System Transfer Function:

Generally, a **transfer function** for a given **linear time-invariant system** is a **mathematical representation**, in **terms of spatial or temporal frequency**, **describing the relation between its input and output assuming zero initial conditions**.

- In order to convert the linear governing relationships to the **frequency domain**, **Laplace transformation** is used.

Example: The transfer function of the cruise control system

-The transfer function of the system output (system velocity v) with respect to the system input (motor voltage u):

$$m\frac{dv(t)}{dt} + cv(t) = ku(t) \xrightarrow{\mathcal{L}\{f(t)\}} mSV(s) + cV(s) = kU(s)$$

$$T(s) = \frac{V(s)}{U(s)} = \frac{k/m}{S + c/m}$$

where the parameter $\boldsymbol{S}$ is a complex number.

-The transfer function of the system output (system velocity v) with respect to the disturbance input (the slope of the road θ) after linearization:

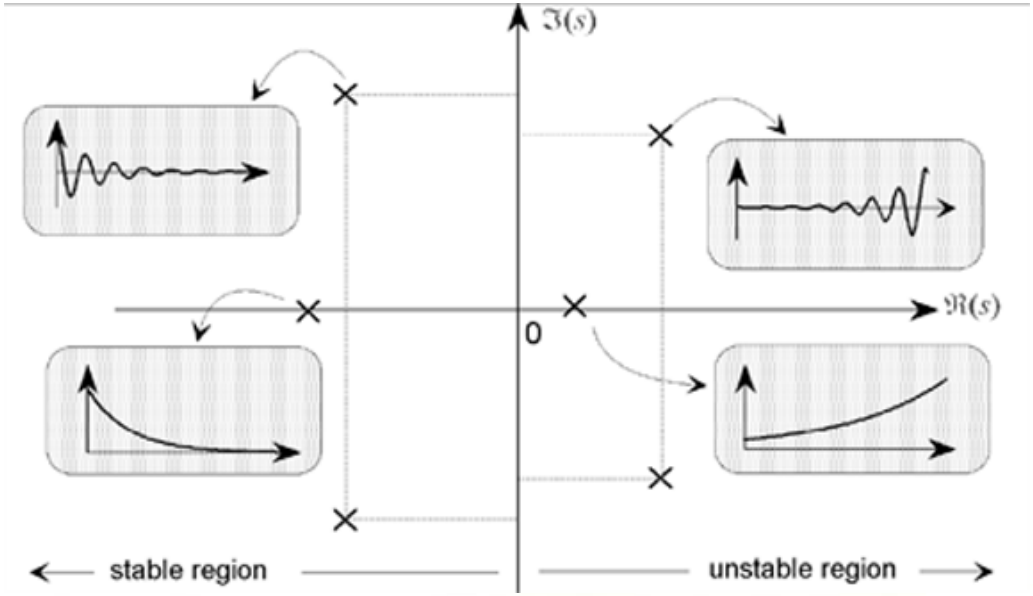
$$m\frac{dv(t)}{dt} + cv(t) = -mg\theta \xrightarrow{\mathcal{L}\{f(t)\}} mSV(s) + cV(s) = -mg\theta(s)$$

$$D(s) = \frac{V(s)}{\theta(s)} = \frac{-g}{S + c/m}$$

System Stability:

A system is defined to be stable if every bounded input to the system results in a bounded output over the time interval $[t_o, \infty)$.

An easy way to identify the stability and the expected response of a system is to observe the poles of its transfer function (the values of ***S*** that makes the denominator become zero) :
When all the poles of the system transfer function are located in the left-half plane (negative real values), the system is shown to be stable.
When one of the poles of the system transfer function is located in the right-half of the S-plane (positive real value), the system becomes unstable.



Transient Response Characteristics:

• The performance characteristics of a control system are frequently specified in terms of the transient response to a step input. The transient response of a practical stable control system often exhibits damped oscillations before reaching steady state value (final value):

* Rise time (t_r): The rise time is the time required for the response to rise from 10% to 90%, 5% to 95%, or 0% to 100% of its final value. In an underdamped system, where the transient response is oscillates within a decaying envelope about the steady state value, the 0% to 100% rise time is normally used. In an overdamped system, where the transient response does not oscillate about the steady state value, the 10% to 90% rise time is commonly used.

* Peak time (t_p): The peak time is the time required for the response to reach the first peak of the overshoot in an underdamped system.

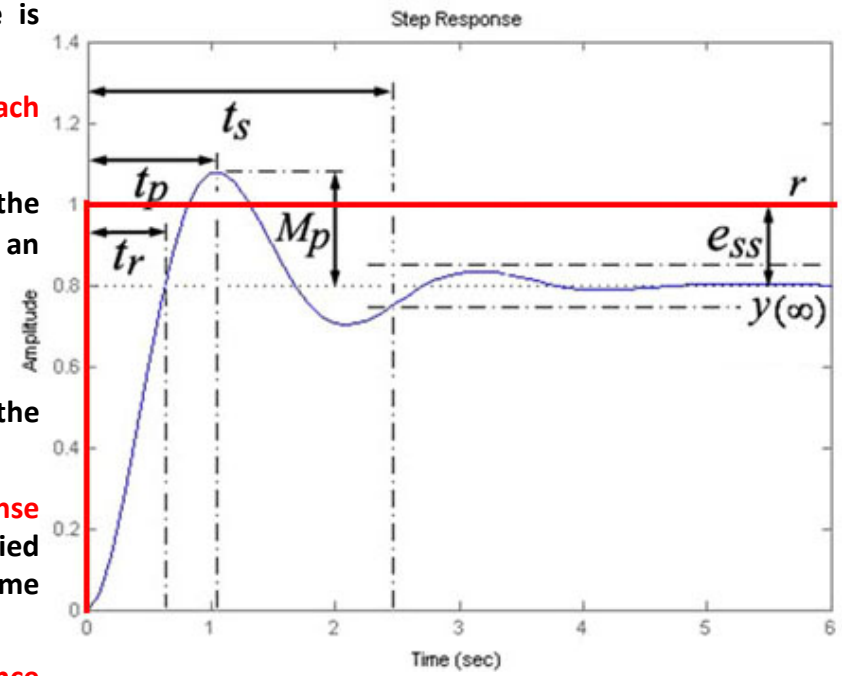
* Maximum (percent) overshoot (M_p): The maximum overshoot is the maximum peak value of the response curve measured from unity in an underdamped system:

$$M_p = \frac{y(t_p) - y(\infty)}{y(\infty)} * 100\%$$

where $y(t_p)$ is the value of the response at the peak time, and $y(\infty)$ is the steady state value.

* Settling time (t_s): The settling time is the time required for the response curve to reach and stay within a range about the final value of sized specified by absolute percentage of the final value (usually 2% or 5%). The settling time is related to the largest time constant of the control system.

* Steady state error (e_{ss}): The steady state error is defined as the difference between the step input to the system and the steady state (final) value of the response: $e_{ss} = \lim_{t\to\infty}(r - y(t))$ where r is the input step to the system.



Proportional-Integral-Derivative (PID) Controller:

- "PID" refers to the three controller correcting calculations, whose sum constitutes the output of the PID controller (the control signal $u(t)$):

* The Proportional part, where the error signal is multiplied by a proportional gain P, is used to eliminate the immediate error (present error).

* The Integral part, where the error signal is integrated (added up) over a period of time and multiplied by an integral gain I, is used to eliminate the steady state error (past error).

* The Derivative part, where the first derivative of the error signal (the slope of the error) is calculated over time and multiplied by a derivative gain D, is used to compensate for future error.

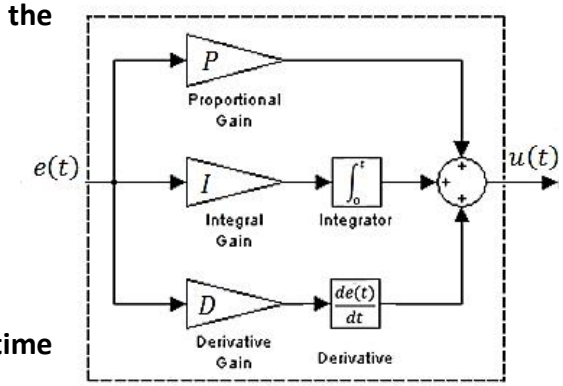
These calculations can be expressed mathematically as follows:

$$u(t) = Pe(t) + I\int_0^t e(\tau).d\tau + D\frac{de(t)}{dt}$$

- In order to discretize the PID controller, an approximation for the integral and the derivative terms can be assumed as following:

$$\int_0^t e(\tau).d\tau = \mathbf{T}\sum_{k=0}^{n} e(k)$$

$$\frac{de(t)}{dt} = \frac{e(n) - e(n-1)}{T}$$



PID Controller

where T is the sampling time period in seconds that should be less than the shortest time constant in the system, and n is the discrete step at time t.

-> Generally, increasing P or I in the controller will decrease the rise time (t_r) of the response, but at the same time will increase the overshoot (M_p) and the settling time (t_s).

-> In addition, increasing D in the controller will decrease the overshoot (M_p) and the settling time (t_s) of the response.

Discrete PID Controller Algorithm :

$$u(t) = Pe(t) + I\int_0^t e(\tau).d\tau + D\frac{de(t)}{dt}$$

- **Position PID Algorithm:** $u(n) = K_p e(n) + K_i T\sum_{k=0}^{n} e(k) + \frac{K_d}{T}(e(n) - e(n-1))$

  Problems: Deferential Kick, Integral Windup, proportional Kick.

Derivative Kick: whenever the reference input is changed in a step form, the error signal will have a corresponding step change as well. The sudden step change of the error signal can cause the derivative of the error signal to be infinite.

In order to eliminate the derivative kick and avoid any unwanted rapid changes in the control signal due to any change in the desired process value, the derivative term can be based on the system output only.

- **Extended Position PID Algorithm:** $u(n) = K_p e(n) + K_i T\sum_{k=0}^{n} e(k) - \frac{K_d}{T}(y(n) - y(n-1))$

  Problems: Integral Windup, proportional Kick.

Discrete PID Controller Algorithm :

$$u(t) = Pe(t) + I\int_0^t e(\tau).d\tau + D\frac{de(t)}{dt}$$

- **Extended Position PID Algorithm:** $u(n) = K_p e(n) + K_i T\sum_{k=0}^{n} e(k) - \frac{K_d}{T}(y(n) - y(n-1))$

  Problems: Integral Windup, proportional Kick.

Integral Windup: In case of larger disturbance than the system capable of handling effects the controlled process, the error between the measured process value (system output) and the reference input value can not be eliminated in a short time. This situation can be problematic due to the integral term of the controller. The integral term will sum up as long as this situation last causing long periods of saturation in the control signal.

In order to eliminate the integral windup, another discrete algorithm can be derived.

- **Velocity PID Algorithm:** $u(n) = u(n-1) + K_p(e(n) - e(n-1)) + K_i T e(n) - \frac{K_d}{T}(y(n) - 2y(n-1) + y(n-2))$

  Problems: proportional Kick.

Discrete PID Controller Algorithm :

$$u(t) = Pe(t) + I\int_0^t e(\tau).d\tau + D\frac{de(t)}{dt}$$

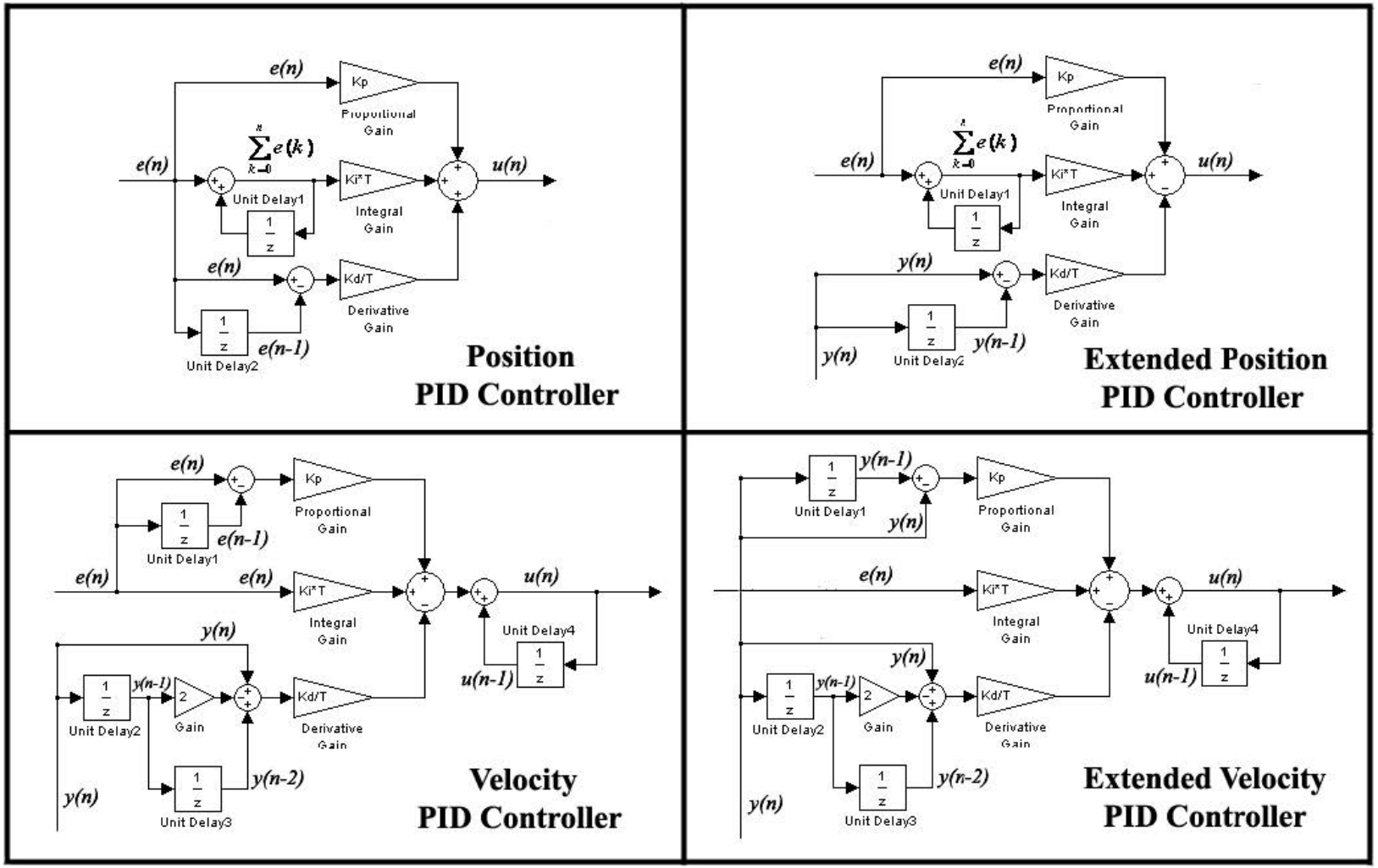
- **Velocity PID Algorithm:** $u(n) = u(n-1) + K_p(e(n) - e(n-1)) + K_i Te(n) - \frac{K_d}{T}(y(n) - 2y(n-1) + y(n-2))$

  Problems: proportional Kick.

Proportional Kick: the proportional term can responds strongly to a step change in the reference input. Although, the proportional kick is much less severe than the derivative kick, it may be too aggressive for some systems.

In order to eliminate the proportional kick, another discrete algorithm can be derived.

- **Extended Velocity PID Algorithm:** $u(n) = u(n-1) + K_p(y(n-1) - y(n)) + K_i Te(n) - \frac{K_d}{T}(y(n) - 2y(n-1) + y(n-2))$

  Problems: Slower Response.

Discrete PID Controller Algorithm :